

Air Conditioners Data Analysis

Stage 1: Data Loading and Reading

This notebook is independently executable in Google Colab. Every step explains what we do, why we do it, what we obtain, and how the result is used in the next stage.

Stage Objective

This stage establishes the starting point of the Air Conditioners analysis. We first load the supplied CSV and verify exactly what data is available before performing any filtering or interpretation.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the project CSV directly so this notebook is independently executable.
csv_path1 = "/content/Most_Expensive_Products.csv"
df1 = pd.read_csv(csv_path1)
csv_path2 = "/content/Top_Discounted_Products.csv"
df2 = pd.read_csv(csv_path2)
csv_path3 = "/content/Top_Rated_Products.csv"
df3 = pd.read_csv(csv_path3)

print("CSV loaded successfully.")
print("Dataset shape:", df1.shape)
print("Dataset shape:", df2.shape)
print("Dataset shape:", df3.shape)
```

```
CSV loaded successfully.
Dataset shape: (15, 14)
Dataset shape: (15, 14)
Dataset shape: (15, 14)
```

```
print('Rows1:', df1.shape[0]); print('Columns1:', df1.shape[1])
print('Rows2:', df2.shape[0]); print('Columns2:', df2.shape[1])
print('Rows3:', df3.shape[0]); print('Columns3:', df3.shape[1])
```

```
Rows1: 15
Columns1: 14
Rows2: 15
Columns2: 14
Rows3: 15
Columns3: 14
```

```
print(df1.columns.tolist())
print(df2.columns.tolist())
print(df3.columns.tolist())
```

```
['name', 'main_category', 'sub_category', 'image', 'link', 'ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', '
['name', 'main_category', 'sub_category', 'image', 'link', 'ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', '
['name', 'main_category', 'sub_category', 'image', 'link', 'ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', ']
```

```
print(df1.dtypes)
print(df2.dtypes)
print(df3.dtypes)
```

```
name                object
main_category       object
sub_category        object
image               object
link                object
ratings             float64
no_of_ratings       float64
discount_price      float64
actual_price        float64
discount_amount     float64
discount_percent    float64
tonnage             float64
energy_star         float64
brand               object
dtype: object
name                object
main_category       object
sub_category        object
image               object
link                object
ratings             float64
no_of_ratings       float64
discount_price      float64
actual_price        float64
discount_amount     float64
discount_percent    float64
tonnage             float64
energy_star         float64
brand               object
dtype: object
name                object
main_category       object
sub_category        object
image               object
link                object
ratings             float64
no_of_ratings       float64
discount_price      float64
actual_price        float64
discount_amount     float64
discount_percent    float64
tonnage             float64
energy_star         float64
```

```
brand          object
dtype: object
```

```
df1.info()
df2.info()
df3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  -
0   name                 15 non-null    object
1   main_category        15 non-null    object
2   sub_category         15 non-null    object
3   image                15 non-null    object
4   link                 15 non-null    object
5   ratings              15 non-null    float64
6   no_of_ratings        15 non-null    float64
7   discount_price       15 non-null    float64
8   actual_price         15 non-null    float64
9   discount_amount      15 non-null    float64
10  discount_percent     15 non-null    float64
11  tonnage              14 non-null    float64
12  energy_star          8 non-null     float64
13  brand                15 non-null    object
dtypes: float64(8), object(6)
memory usage: 1.8+ KB
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  -
0   name                 15 non-null    object
1   main_category        15 non-null    object
2   sub_category         15 non-null    object
3   image                15 non-null    object
4   link                 15 non-null    object
5   ratings              15 non-null    float64
6   no_of_ratings        15 non-null    float64
7   discount_price       15 non-null    float64
8   actual_price         15 non-null    float64
9   discount_amount      15 non-null    float64
10  discount_percent     15 non-null    float64
11  tonnage              10 non-null    float64
12  energy_star          7 non-null     float64
13  brand                15 non-null    object
dtypes: float64(8), object(6)
memory usage: 1.8+ KB
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  -
0   name                 15 non-null    object
1   main_category        15 non-null    object
2   sub_category         15 non-null    object
3   image                15 non-null    object
4   link                 15 non-null    object
5   ratings              15 non-null    float64
6   no_of_ratings        15 non-null    float64
7   discount_price       15 non-null    float64
8   actual_price         15 non-null    float64
9   discount_amount      15 non-null    float64
10  discount_percent     15 non-null    float64
```

```
print(df1.isna().sum())
print(df2.isna().sum())
print(df3.isna().sum())
```

```
name          0
main_category 0
sub_category   0
image          0
link           0
ratings        0
no_of_ratings  0
discount_price 0
actual_price   0
discount_amount 0
discount_percent 0
tonnage        1
energy_star    7
brand          0
dtype: int64
name          0
main_category 0
sub_category   0
image          0
link           0
ratings        0
no_of_ratings  0
discount_price 0
actual_price   0
discount_amount 0
discount_percent 0
tonnage        5
energy_star    8
brand          0
dtype: int64
name          0
main_category 0
sub_category   0
image          0
link           0
ratings        0
no_of_ratings  0
discount_price 0
actual_price   0
discount_amount 0
discount_percent 0
tonnage        1
energy_star    3
```

```
brand          0
dtype: int64
```

```
print('Total missing values1:', int(df1.isna().sum().sum()))
print('Total missing values2:', int(df2.isna().sum().sum()))
print('Total missing values3:', int(df3.isna().sum().sum()))
```

```
Total missing values1: 8
Total missing values2: 13
Total missing values3: 4
```

```
print('Duplicate rows1:', int(df1.duplicated().sum()))
print('Duplicate rows2:', int(df2.duplicated().sum()))
print('Duplicate rows3:', int(df3.duplicated().sum()))
```

```
Duplicate rows1: 0
Duplicate rows2: 0
Duplicate rows3: 0
```

```
print(df1.describe(include='all').T)
print(df2.describe(include='all').T)
print(df3.describe(include='all').T)
```

```
count unique \
name          15    15
main_category 15     1
sub_category  15     1
image         15    14
link          15    15
ratings       15.0   NaN
no_of_ratings 15.0   NaN
discount_price 15.0   NaN
actual_price  15.0   NaN
discount_amount 15.0   NaN
discount_percent 15.0   NaN
tonnage       14.0   NaN
energy_star   8.0    NaN
brand         15     6
```

```
top freq \
name          Voltas 3Ph Scroll SAC C2-Nventure I-Cassette A... 1
main_category          appliances 15
sub_category          Air Conditioners 15
image  https://m.media-amazon.com/images/I/611TN01YAg... 2
link  https://www.amazon.in/Voltas-Scroll-SAC-C2-Nve... 1
ratings          NaN NaN
no_of_ratings          NaN NaN
discount_price          NaN NaN
actual_price          NaN NaN
discount_amount          NaN NaN
discount_percent          NaN NaN
tonnage          NaN NaN
energy_star          NaN NaN
brand          Voltas 6
```

```
mean      std      min      25%      50% \
name      NaN      NaN      NaN      NaN      NaN
main_category      NaN      NaN      NaN      NaN      NaN
sub_category      NaN      NaN      NaN      NaN      NaN
image      NaN      NaN      NaN      NaN      NaN
link      NaN      NaN      NaN      NaN      NaN
ratings    3.706667    1.032657    1.0    3.3    3.8
no_of_ratings    7.6    6.652604    1.0    2.5    5.0
discount_price  85027.333333  14918.057355  71990.0  77190.0  79350.0
actual_price    97288.0  20807.210014  77900.0  84130.0  89470.0
discount_amount  12260.666667  8880.813562  2000.0  3650.0  11980.0
discount_percent  11.734283    7.310246  2.567394  4.39759  13.389963
tonnage    2.314286    0.601098    2.0    2.0    2.0
energy_star    3.625    1.187735    2.0    3.0    3.0
brand      NaN      NaN      NaN      NaN      NaN
```

```
75%      max
name      NaN      NaN
main_category      NaN      NaN
sub_category      NaN      NaN
image      NaN      NaN
link      NaN      NaN
ratings    4.1    5.0
no_of_ratings    13.5    18.0
discount_price  85030.0  128800.0
actual_price  103950.0  149000.0
```

```
print('Stage 1 baseline established.')
```

```
Stage 1 baseline established.
```